# MD TARIF RAIHAN

EXPERIMENTAL THERMAL & ENERGY ENGINEERING

Milwaukee, Wisconsin | mdraihan@uwm.edu | github.com/tarif-raihan | Google Scholar

Mechanical Engineering Ph.D. candidate (expected May 2027) and Certified Energy Manager. I build instrumented thermal-fluid experiments, write the physics models that explain them, and use validated models for control: transpiration cooling through sintered metal plates, and liquid cooling for high-heat-flux electronics.

## RESEARCH & PROFESSIONAL EXPERIENCE

### Graduate Research Assistant | UW-Milwaukee | Aug 2022-Present

Department of Mechanical Engineering; advisor: Prof. Ryoichi S. Amano.

- Designed and ran transpiration-cooling experiments on sintered SS316L and titanium porous plates (wire-mesh and powder) in a channel facility (hydraulic diameter 0.14 m) at Reynolds numbers of 39,000-70,000; titanium performed better at every condition (GT2025-151170).
- Wrote 1-D and 2-D Python local-thermal-non-equilibrium Darcy-Forchheimer solvers, validated them against the experiments, and corrected a Kozeny-Carman permeability mismatch. Sensitivity analysis ranked solid conductivity first, pore architecture second, pore diameter last (ASME JERT, 2026).
- Measured a thermal time constant of 48 +/- 2.3 s for titanium plates on a closed-loop aerothermal platform with PWM fan control, and trained a neural network on the measured data to predict surface temperature for control (GT2026; HT2026).
- Designing a bench-scale data-center liquid-cooling test vehicle (cartridge-heated copper block, LabJack T7-Pro, NVIDIA Jetson edge control) with a physics-informed neural-network surrogate trained on 10,000 simulated operating points. Hardware validation in progress.
- Wrote the full proposal for, and lead, a $50,000 UWM Research Foundation Catalyst project: graded versus uniform porous cold plates at equal pumping power, target 200 W/cm2, industry partners Eaton and RPI Inc.

### Industrial Energy Assessor & Report Reviewer | U.S. DOE ITAC, UW-Milwaukee | 2022-Present

- Assessed 36 manufacturing plants and led 14 assessments, with about 8-9 recommendations per report on HVAC, compressed air, motors, lighting, and waste-heat recovery.
- Reviewed 38 ITAC and BTAC reports for assumptions, savings calculations, and feasibility. Trained undergraduate assessors.

### Scientific Computing Research | NSF CIBERCATSS | Summer 2026

JAX-FEM differentiable inverse modeling of porous media (mentor: Dr. Tian Zhao). Showed porosity is identifiable from temperature data but permeability is not, which sets the sensors future tests need.

### Engineering Intern | Clean Energy Program, RPI Inc., Racine, WI | Aug 2023-Jun 2024

- Waste-heat recovery and thermal-system techno-economic analyses.
- Designed a PLC-board clamp for ABB in SolidWorks (about 500 units manufactured) and a magnetic connector for Nordco.

### Graduate Researcher (M.Sc. Thesis) | Technische Hochschule Ingolstadt, Germany | 2022

MATLAB/Simulink (CARNOT) model of a hybrid air- and ground-source heat pump. Cut required borehole depth by 21% (70 m to 55 m); at equal depth, the switching strategy alone raised the annual performance factor by 15% (2.98 to 3.42).

### Energy Engineer | Stichting Bangladesh Accord Foundation | Apr 2018-Jun 2020

Inspected 467 garment factories for fire and building safety; wrote corrective action plans and followed them to closure.

### Sustainable Infrastructure Development Engineer | Fabulous Fashions Ltd., Bangladesh | Nov 2014-Apr 2018

Energy-efficiency projects under ISO 50001, ISO 14001, and LEED v4, including boiler and generator efficiency tests and a solar PV system supplying 10% of plant energy. Led a team of 12 engineers and technicians.

## TEACHING & MENTORING

### Teaching Assistant | UW-Milwaukee | Fall 2022-Present

- ME 320 Introduction to Fluid Mechanics (classes of up to 86): led discussion sections, delivered a lecture, and ran a pipe pressure-drop laboratory module. Student ratings 4.56/5 and 4.83/5. Also supported ME 321, ME 360, and ME 430.
- Mentored seven undergraduate researchers (SURF); trained four undergraduates in industrial energy assessment; guided 11 students on plant-recommendation projects.

## EDUCATION & CERTIFICATION

**Ph.D., Mechanical Engineering** | University of Wisconsin-Milwaukee | expected May 2027

**M.Sc., Renewable Energy Systems** | Technische Hochschule Ingolstadt, Germany | 2022

**B.Sc., Mechanical Engineering** | Chittagong University of Engineering and Technology | 2014

**Certified Energy Manager (CEM)** | Association of Energy Engineers | 2025

## TECHNICAL EXPERTISE

**Experimental:** transpiration and convective heat-transfer testing; test rig design; thermocouple, flow, and pressure-drop instrumentation; LabJack data acquisition; PWM control; NVIDIA Jetson edge computing; calibration and uncertainty analysis.

**Computational:** Python; MATLAB/Simulink; CARNOT; JAX/JAX-FEM; LTNE Darcy-Forchheimer modeling; physics-informed neural networks; ANSYS; COMSOL; SolidWorks.

**Energy:** industrial energy assessment; compressed-air, HVAC, and motor systems; heat pumps; waste-heat recovery; ISO 50001.

## SELECTED PUBLICATIONS

Raihan, M. T., and Amano, R. S. (2026). Decoupling the roles of conductivity, architecture, and pore diameter in transpiration cooling through sintered metallic porous media. ASME J. Energy Resources Technology, Part A. doi:10.1115/1.4072749.

Raihan, M. T., and Amano, R. S. (2026). Performance analysis and optimization of a dual-source heat pump system: Evaluating bivalent point strategies for enhanced efficiency. Int. J. Energy Efficiency Engineering, 1(2).

Raihan, M. T., and Amano, R. S. (2026). Artificial intelligence-driven thermal control in advanced transpiration cooling systems. ASME Turbo Expo, GT2026-176271. doi:10.1115/GT2026-176271.

Raihan, M. T., and Amano, R. S. (2025). Transpiration cooling in gas turbine trailing edge using stainless steel and titanium porous media. ASME Turbo Expo, GT2025-151170. doi:10.1115/GT2025-151170.

Raihan, M. T., and Amano, R. S. (2025). Enhanced transpiration cooling for aerospace systems: A comparative study of sintered wire mesh vs. sintered powder materials. ASME HT2025-156087. doi:10.1115/HT2025-156087.

Raihan, M. T., and Amano, R. S. (2025). Enhanced transpiration cooling for aerospace systems: A comparative study of stainless-steel wire mesh with 80 and 100-micron pores. ASME IMECE2025-164512. doi:10.1115/IMECE2025-164512.

Amano, R. S., and Raihan, M. T. (2025). Artificial intelligence as a new frontier in turbomachinery design and applications. In Recent Developments in Power and Propulsion Applications, pp. 641-662. Springer. doi:10.1007/978-981-96-8602-5_21.

Full list (5 journal articles, 6 ASME conference papers, 1 book chapter, 2 manuscripts in review): see the Publications page.

## AWARDS & SERVICE

ME Graduate Students Research Achievement Award, UWM (2026); NSF CIBERCATSS selected participant (2026); Chancellor's Graduate Student Award, UWM (2025); ME Graduate Students Research Achievement Award, UWM (2024); Finalist, Michael Krauski Memorial Award (2024). President, Society of Power Engineers, UWM (2024-present). Member: ASME, ASHRAE, AEE.